

Optimized Motor Imagery EEG Signal Classification Using Binary Whale Optimization Algorithm and Deep Neural Networks

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Abstract— Classifying EEG signals for Motor Imagery (MI) is a key step in building effective Brain–Computer Interface (BCI) systems. It is not easy to extract features from EEG due to high dimensionality and overlapping features and this makes classification a challenging task. To address this, we propose a hybrid approach that combines Binary Whale Optimization Algorithm (BWOA) for selecting relevant features with a Deep Neural Network (DNN) for classification. After standard preprocessing, features are extracted using time-frequency analysis and autoregressive (AR) modelling. BWOA then selects the most meaningful features by applying a binary mask that filters out irrelevant data. The selected features are used to train a four-layer DNN, optimized with the Adam algorithm and validated through cross-validation. To strike a balance between accuracy and simplicity, we also design a custom fitness function for BWOA that considers both classification performance and feature reduction. The results clearly indicate the superiority of the proposed method over the state-of-the-art methods.

Keywords— EEG, DNN and BCI

I. INTRODUCTION

Brain–Computer Interfaces (BCIs) are emerging as a key technology for enabling direct communication between the brain and external devices, with significant applications in assistive systems for individuals with severe motor disabilities [1]. One of the most promising techniques in this domain is Motor Imagery (MI), where a person imagines moving a part of their body [2]. MI has gained considerable attention due to its natural interaction style and its ability to consistently generate distinct brain signals that can be effectively recognized and analyzed by BCI systems [3]. The most popular and practical method for recording brain activity in BCI systems is electroencephalography (EEG). It's widely used because it's non-invasive, relatively affordable, and capable of capturing brain signals in real time. However, EEG data is corrupted with external noises and can vary with body movement. As a result, the effectiveness of any BCI system depends heavily on how well the data is pre-processed, which features are extracted, and how those features are classified.

BCI systems operate in two parts. The first part is the offline training phase, where system learning is performed from pre-recorded EEG data using machine learning models. The second phase is online, where system in real time reads the live EEG signals and translates them into commands to control an external device.

Over the years, researchers have explored many techniques to extract useful information from EEG signals. These include time-domain features, signal complexity measures like entropy, and even hybrid approaches that

combine multiple signal characteristics [4]. Some studies—such as one by Gunes et al. have even shown that combining EEG with other physiological signals, like eye movements, can significantly boost classification accuracy, especially in applications like sleep monitoring.

To interpret these features effectively, researchers have experimented with a wide range of classification models. Approaches like Multilayer Perceptrons (MLPs), Deep Neural Networks (DNNs), and Long Short-Term Memory (LSTM) networks have all shown potential. LSTMs excel at capturing temporal patterns in EEG data, while DNNs are often preferred for being fast, lightweight, and easy to train. Still, regardless of the model, the quality of input features remains a critical factor [5]. Too many irrelevant or overlapping features can actually reduce a model's performance.

To address this issue, our study proposes a novel approach that combines a DNN with the Binary Whale Optimization Algorithm (BWOA) [6] for feature selection. By using BWOA to filter out less relevant features, the DNN can focus on the most informative inputs—resulting in improved classification accuracy and more reliable performance. This hybrid approach aims to build BCI systems that are not only more accurate, but also efficient and practical for real-time use.

II. LITERATURE SURVEY

Classifying MI using EEG signals has long been a central focus in the development of BCI systems. Early research primarily relied on segmenting EEG signals and applying **traditional** classification techniques. For example, Hsu [7] proposed an improved method for identifying the most active signal segments, which were then paired with adaptive classifiers to enhance performance in MI-related tasks. Classic machine learning models such as Support Vector Machines (SVMs) and statistical neural networks were widely explored during this period. Übeyli [8] employed eigenvector-based methods in combination with multiclass SVMs for more effective EEG signal analysis, while Güler et al. [9] utilized recurrent neural networks and Lyapunov exponents to better capture the temporal dynamics of EEG data. Some of the foundational theoretical work on neural networks—particularly from a statistical standpoint—was laid by Titterton and Cheng [10], helping establish a basis for the deep learning approaches used today. As deep learning technologies advanced, researchers began applying them more extensively to EEG classification. Craik et al. [11] explored the use of deep learning models to uncover and learn complex patterns within EEG data. Schirmeister et al. [12] demonstrated the use of Convolutional Neural Networks

(CNNs) to not only classify but also visualize EEG activity. Similarly, Jirayucharoensak et al. [13] incorporated covariate shift adaptation into deep networks for emotion recognition from EEG signals. Zhang et al. [14] introduced a deep attention-based LSTM model specifically designed for classifying hand movements, highlighting the growing importance of temporal modelling in MI classification. Further improvements in preprocessing and optimization have also contributed to the enhanced performance of EEG-based systems. For instance, Bhateja et al. [15] applied advanced anisotropic diffusion techniques in image processing, which offer parallels to EEG signal enhancement methods. Additionally, Sobol [16] introduced global sensitivity analysis techniques that can be adapted to evaluate feature importance in EEG models.

Together, these **contributions** reflect a clear shift from traditional methods toward more powerful, adaptive, and hybrid systems. This progression has inspired the use of metaheuristic algorithms for feature selection in combination with DNNs, forming the basis of the approach proposed in this study.

A. EEG Preliminaries

EEG can be defined as a signal that captures human brain activity using electrodes placed on the scalp (Fig. 1). The specific channels selected for data acquisition are also described in detail.

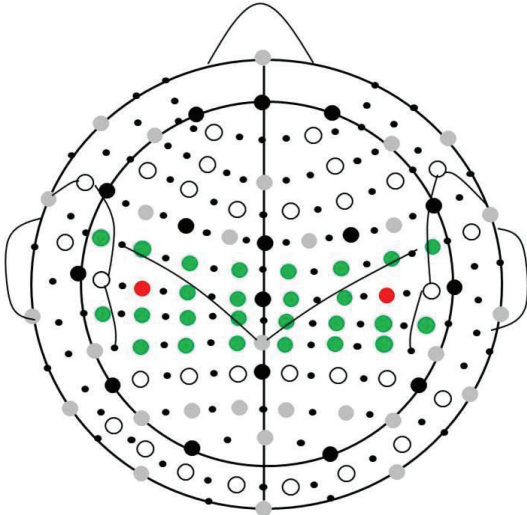


Fig. 1. 10-20 System-Based Configuration for 128 EEG Electrodes

In motor imagery-based BCI systems, the relevant EEG activity typically falls within the 8–30 Hz range, which is commonly segmented into two distinct frequency bands: the alpha band (8–13 Hz) and the beta band (14–30 Hz).

III. PROPOSED WORK

Fig. 2 shows the **block** diagram of the proposed method.

Typically, there are mainly 3 stages in MI EEG signal classification. The first stage focuses on preparing the data, which **includes** steps like signal decomposition, artifact removal, and noise reduction. In the second stage, we have extraction of meaningful features from the preprocessed signals. Finally, in the third stage, these features are used to classify the EEG signals into relevant MI categories.

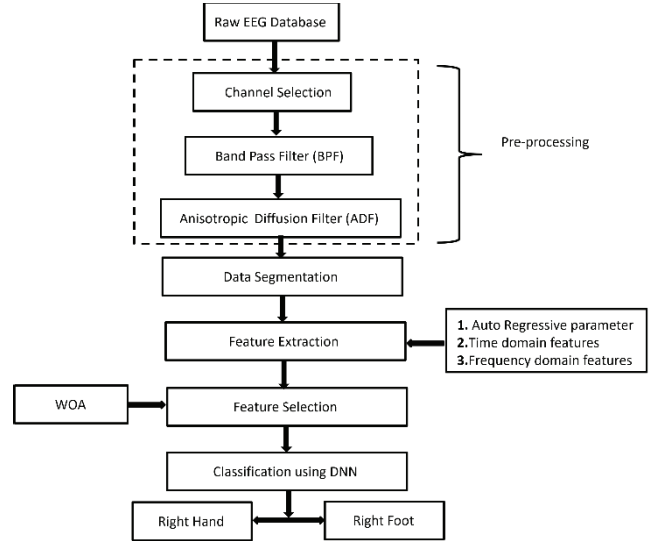


Fig. 2. Block diagram of proposed method

A. Channel selection

EEG recordings typically involve signals from 118 channels, which makes direct analysis of the full dataset highly complex and **computationally** intensive. To manage this, the study focuses on a two-class MI task involving right-hand and right-foot movements, a subset of 30 EEG channels located over the motor cortex is selected for analysis, as illustrated in Fig. 1.

B. Band Pass Filtering

To obtain an EEG signal free from human-induced artifacts and confined **within** the relevant frequency range of 7–30 Hz, a band-pass Butterworth filter can be applied. This filter effectively removes noise and unwanted frequency components while preserving the signal of interest.

$$H(x, y) = 1 - \frac{1}{1 + \left(\frac{D(x, y)\Omega}{D^2(x, y) - D_0^2} \right)^{2n}} \quad (1)$$

In the above expression, the term $D(x, y)$ denotes the distance from a specific point in the frequency domain to the filter's center frequency. The variable Ω represents the bandwidth of the filter, while n refers to the order of the Butterworth filter, which controls the sharpness of the transition between the passband and stopband.

C. Anisotropic Diffusion Filtering (ADF)

In this study, ADF is adapted for EEG signal denoising, aiming to reduce unwanted noise without compromising key signal features relevant for MI analysis.

D. Data Segmentation

Following preprocessing, the next step involves segmenting the data. Each EEG channel undergoes 168 trials, resulting in a dataset with dimensions of 168×30 , where 30 corresponds to the selected EEG channels. Among the total trials, 80 correspond to right-hand MI and 88 to right-foot imagery. For the purpose of **model** development, the dataset is split into three subsets: 100 trials are used for training, 34 for validation, and the remaining 34 trials are reserved for testing the performance of the classification model.

E. Feature Extraction

In MI classification, feature extraction plays a vital role. The features extracted need to effectively distinguish between

different classes to ensure accurate classification. To achieve this, features **from** multiple domains as discussed below are taken into account.

1) Analysis of the Time Domain

In the beginning, the EEG signal is represented in the time domain, where its statistical properties are analysed. Specifically, the value of the signal's sub-mean band is estimated by:

$$m = \frac{1}{N} \sum_{i=1}^N E_i \quad (2)$$

The standard **deviation** of the EEG signal is computed using the following formula

$$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (E_i - m)^2} \quad (3)$$

Zero crossing represents the number of times an EEG signal goes beyond the zero line—that is, when the signal changes from positive to negative or vice versa. To reduce the impact of noise and artifacts, a small tolerance band around zero is typically applied during this calculation

$$\{E_i > 0 \text{ and } E_{i+1} < 0\} \text{ or } \{E_i < 0 \text{ and } E_{i+1} > 0\} \\ \text{And } |E_i - E_{i+1}| \geq \varepsilon \quad (4)$$

2) Frequency Domain Analysis

In frequency domain analysis, signals are characterized by examining their frequency **components**, which are obtained by applying the discrete Fourier transform (DFT) to the signal $E(n)$, defined as follows:

$$E(k) = \sum_{n=0}^{N-1} E(n) e^{-i\frac{2\pi}{N}kn} \quad (5)$$

a) Band Power

The power band refers to the signal's power within a specific frequency range. Since EEG signals are inherently non-stationary, linear time-invariant (LTI) system assumptions cannot be directly applied to power measurements in a single band. Instead, the power spectral density (PSD) of the signal is calculated to provide a more accurate representation of power distribution across frequencies, using the following expression:

$$S(k) = \left| \sum_{n=0}^{N-1} E(n) e^{-i\frac{2\pi}{N}kn} \right|^2 \quad (6)$$

To estimate the power within a specific frequency band, the time-domain signal is divided into blocks. The periodogram for the M^{th} block is calculated using the following formula:

$$P_M(k) = \frac{1}{M} \left| \sum_{n=0}^{N-1} E(n) e^{-i\frac{2\pi}{N}kn} \right|^2 \quad (7)$$

3) Autoregressive parameter

The autoregressive (AR) model is a powerful tool for representing time series data. the power spectral density (PSD) can be accurately estimated as:

$$p_x(e^{j\omega}) = \frac{\hat{E}N_k}{|(1 + \sum_{k=1}^p a_k e^{j\omega k})|^2} \quad (8)$$

The coefficients a_i 's are evaluated as

$$\|a\|_p = (\sum_{i=1}^n |a_i|^p)^{1/p} \quad (9)$$

Consequently, each trial in both the training and testing phases produces '6' features. This results in a dataset of size $148 \times 30 \times 6$, where 100 trials by 30 channels by 6 features are allocated for training, 20 trials by 30 channels by 6 features for validation, and the remaining 48 trials by 30 channels by 6 features are reserved for testing.

F. Feature Selection

The feature selection process is an important criterion in EEG classification process. Let us define N as the number of EEG samples (trials) and D is total number of extracted features (TF + AR). Further defining the EEG **feature** matrix as

$$F = \begin{pmatrix} f_{11} & f_{12} & \dots & f_{1D} \\ f_{21} & f_{22} & \dots & f_{2D} \\ \vdots & \vdots & \ddots & \vdots \\ f_{N1} & f_{N2} & \dots & f_{ND} \end{pmatrix} \in R^{N \times D} \quad (10)$$

1) Binary Selection Vector

Each whale represents a binary vector as

$$W = [w_1, w_2, \dots, w_D] \quad w_j \in \{0, 1\} \quad (11)$$

If $w_j = 1$, then feature j is selected, and if $w_j = 0$, then feature j is discarded.

The selected feature **matrix** is:

$$F^{(sel)} = F \odot W \quad (12)$$

where $\odot$ is the element-wise (Hadamard) product that multiplies each feature by 0 or 1.

2) Fitness Function using DNN

The fitness function evaluates how good a feature subset is by training a DNN classifier on $F^{(sel)}$ and computing classification performance. The goal is to minimize both classification error and the number of features, using the fitness function as

$$F_{Fit}(W) = \alpha \cdot (1 - A_{CCDNN}(W)) + \beta \frac{|W|}{D} \quad (13)$$

where, $A_{CCDNN}(W)$ is the accuracy of the DNN classifier trained on selected features, $|W| = \sum_{j=1}^D w_j$ denotes the number of selected features and α, β are weighting coefficients, such that $\alpha + \beta = 1$.

3) Whale Optimization in Binary Search Space

Various algorithms based on nature have been introduced recently for solving complex optimization issues, especially in high-dimensional and non-linear search spaces. Among these, the WOA stands out due to its simplicity, flexibility, and strong global search capability. However, many real-world problems, such as feature selection, require searching in binary search spaces, where solutions are represented as sequences of 0s and 1s. To address this, the Whale Optimization Algorithm has been adapted into a BWOA, where position updates are guided by probabilistic transfer functions that map continuous updates into binary decisions. This adaptation allows WOA to be applied effectively to discrete problems, opening new opportunities in fields such as machine learning, pattern recognition, and combinatorial optimization.

Whales assume that the current best solution is the prey (i.e., the target), and they move closer to it. The update equations are:

$$\vec{D} = |\vec{C} \cdot \vec{W}^*(t) - \vec{W}(t)| \quad (14)$$

Another movement strategy is to simulate the spiral-shaped path of whales as they encircle their prey. When $|\vec{A}| \geq 1$, the whale explores the search space randomly by moving toward a random whale $\vec{W}_{rand}(t)$

$$\vec{W}(t+1) = \begin{cases} \vec{W}^*(t) - \vec{A} \cdot \vec{D} & \text{if } p < 0.5, |\vec{A}| < 1 \\ \vec{W}_{rand}(t) - \vec{A} \cdot \vec{D} & \text{if } p < 0.5, |\vec{A}| \geq 1 \\ \vec{D} \cdot e^{bl} \cdot \cos(2\pi l) + \vec{W}^*(t) & \text{if } p \geq 0.5 \end{cases} \quad (15)$$

where, $\vec{W}^*(t)$ is best whale position so far, $\vec{W}_{rand}(t)$ is the random whale, a linearly decreases from 2 to 0. Further

$$A = 2a \cdot r_1 - a, C = 2 \cdot r_2, r_1, r_2 \in [0,1] \quad (16)$$

WOA's continuous behaviour is mapped to binary space using a sigmoid transfer function.

$$S(w) = \frac{1}{1+e^{-w}} \quad (17)$$

$$w_j(t+1) = \begin{cases} 1, & \text{if } rand < S(w_j(t+1)) \\ 0, & \text{otherwise} \end{cases} \quad (18)$$

G. Classification using DNNs

The architecture of the DNN is designed to balance representational capacity and generalization, especially given the reduced input dimensionality after feature selection.

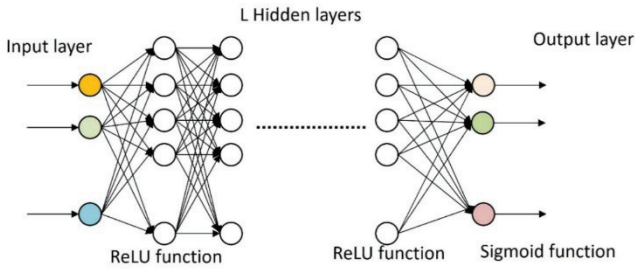


Fig. 3. DDN structure for Classification

A typical DNN structure for EEG signal classification includes the following layers (Fig. 3). The input layer receives the selected EEG feature vector. Input dimensionality equals the number of selected features, denoted as:

$$\text{Input size} = |W| = \sum_{j=1}^D w_j \quad (19)$$

where $W \in \{0,1\}^D$ is the binary feature selection mask from BWOA. In this work, 4 hidden layers are considered, and non-linearity is introduced using Rectified Linear Unit (ReLU) defined as

$$f(z) = \max(0, z) \quad (20)$$

A final dense layer with neurons equal to the number of target classes C . Softmax activation is applied to produce class probabilities as

$$\hat{y}_i = \frac{e^{z_j}}{\sum_{k=1}^C e^{z_k}} \quad \forall j \in \{1, 2, \dots, C\} \quad (21)$$

The loss function used is categorical cross-entropy, defined as:

$$L(y, \hat{y}) = -\sum_{i=1}^C y_i \cdot \log(\hat{y}_i) \quad (22)$$

where $y_i \in \{0,1\}$ is the true label (one-hot encoded), and $\hat{y}_i$ is the predicted probability.

To enhance robustness and reduce the risk of overfitting, a 5-fold cross-validation strategy is employed.

The DNN accuracy is computed as:

$$A_{CCDNN} = \frac{1}{k} \sum_{i=1}^k \frac{TP_i + TN_i}{TP_i + TN_i + FP_i + FN_i}$$

IV. RESULTS

This section is divided into two parts to better explain the workflow and findings of the study. The first part outlines the core steps involved in processing the EEG signals. This includes how the data was cleaned, structured, and prepared using a standardized electrode placement system adapted for 128 channels. Special attention was given to ensuring the signals were free from noise and ready for meaningful analysis. Various techniques were applied to highlight important features in the data that could help distinguish between different brain activity patterns.

The second part focuses on the classification approach used to interpret the processed EEG signals. Simulations were carried out to test how well the proposed method could recognize specific patterns—particularly those related to motor imagery tasks. The goal here was to build a reliable system that could learn from the data and make accurate predictions. Throughout this phase, different models and validation strategies were applied to evaluate performance and ensure the approach could be generalized to other similar tasks.

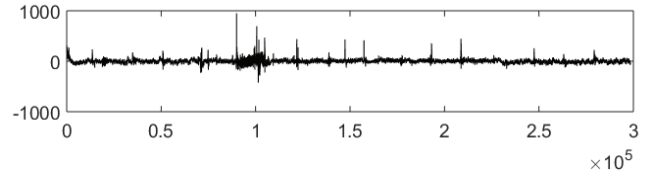


Fig. 4. Recorded EEG signal

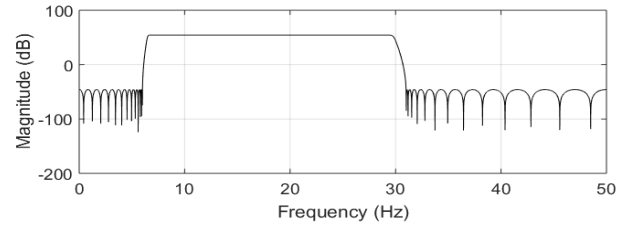


Fig. 5. Magnitude spectrum of EEG signal

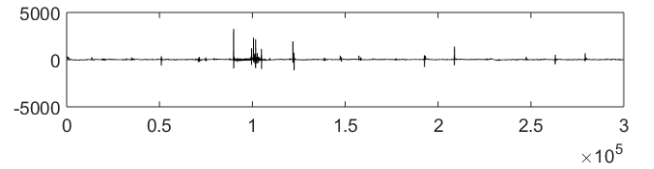


Fig. 6. Filtered EEG signal

Fig. 5 illustrates the raw EEG signal recorded during the MI experiment. As seen, the signal contains a mix of relevant brain activity along with background noise and artifacts.

Fig. 6 presents the corresponding magnitude spectrum of the EEG signal, highlighting the frequency components and revealing dominant peaks within the typical EEG bands such as alpha and beta. To enhance signal clarity, filtering techniques were applied to remove noise and retain meaningful components, as shown in Fig. 6, which displays the cleaned EEG signal with significantly improved waveform smoothness and reduced high-frequency artifacts.

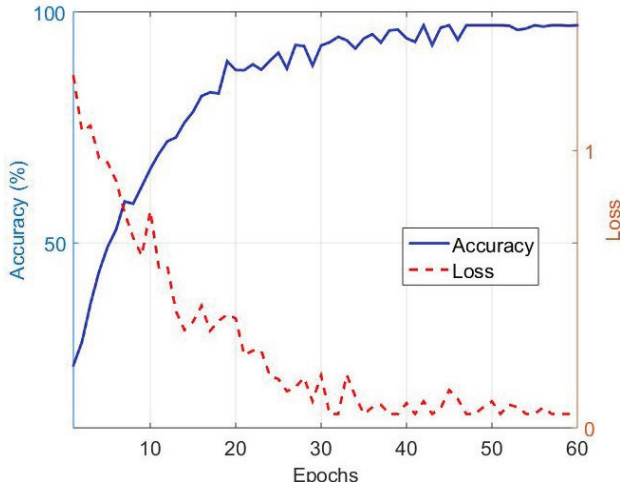


Fig. 7. Loss and Accuracy Curve

Fig. 7 presents the training loss and accuracy curves over 60 epochs for the deep neural network classifier. The accuracy curve shows a steady improvement from an initial value of approximately 14%, eventually converging to a peak value of 97.11%, indicating effective learning and model generalization. Simultaneously, the loss curve exhibits a consistent decline with mild fluctuations, decreasing from an initial high value and stabilizing around 0.05. The overall trends in both curves reflect successful convergence of the network and effective optimization during training.

TABLE I. CLASSIFICATION ACCURACY COMPARISON

Author	Algorithm	Accuracy%
Aboalayon et al., 2018, [17]	DL-CNN	92.2
Zhou et. al. 2018, [18]	LSTM	91.4
Amin et.al, 2019, [19]	DL-CNN	95.4
Kumar et.al, 2019, [20]	LSTM	95.5
Fadel et.al., 2020, [21]	CNN-LSTM	70.64
Zheng et.al., 2019, [14]	DNN	67.76
Nakra et.al., 2019, [22]	DNN	95.34
Proposed	DNN+ WOA	97.11

Recent studies focus on deep learning methods for better classification in BCI systems. For example, Aboalayon et al. [17] used a CNN and achieved a good 92.2% accuracy. Zhou et al. [18] considered LSTM network for classification and reached 91.4% accuracy. Other studies, such as those by Amin et al. [19] and Kumar et al. [20], reported strong results with accuracies of 95.4% and 95.5% using DL techniques.

Fadel et al. [21] used a hybrid CNN-LSTM model but only reached 70.64% accuracy. This highlights some of the challenges in effectively combining different deep learning architectures. Similarly, Zheng et al. [14] used a DNN and reported 67.76% accuracy. This suggests that factors like data quality, preprocessing steps, and how features are represented can greatly impact results. On the other hand, Nakra et al. [22] demonstrated how effective a well-tuned deep learning model, and achieve an accuracy of 95.34%. In comparison, our approach, which pairs a DNN with feature selection using the BWOA, reached a much higher accuracy of 97.11%. This clear improvement shows how combining nature-inspired optimization with deep learning can really help—especially by focusing on the most important features and cutting out unnecessary data.

V. CONCLUSION

This research article introduced a practical and useful framework for classifying MI- EEG signals. The approach combines time-frequency and autoregressive feature extraction with BWOA for feature selection, followed by classification using a DNN. By applying BWOA, we were able to reduce the feature space efficiently, selecting only the most relevant features while maintaining high classification performance. The DNN, designed with four hidden layers, was trained on this optimized feature set and showed strong learning behaviour—evident from the steady rise in accuracy and consistent drop in loss across 60 training epochs. The proposed technique is quite useful as it has a high classification accuracy of 97.11%. These results highlight the strength of combining evolutionary optimization with deep learning, making the framework a promising solution for building accurate and scalable BCI systems. In future work, we aim to explore real-time deployment and cross-subject adaptability to assess its performance in real-world BCI applications.

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